to the 15-factor methodology. The attachment is done through resource binding, which in the case of PostgreSQL, consists of the following:

- A URL to define which driver to use, where to find the database server, and which database to connect the application to
- Username and password to establish a connection with the specified database

Thanks to Spring Boot, you can provide those values as configuration properties. This means you can easily replace the attached database by changing the values for the resource binding.

Open the application.yml file for the Catalog Service project, and add the properties for configuring the connection with PostgreSQL. Those values are the ones you defined earlier as environment variables when creating the PostgreSQL container.

**Listing 5.2 Configuring the connection to a database using JDBC**

```
spring:
  datasource:                    ❶
    username: user
    password: password
    url: jdbc:postgresql://localhost:5432/polardb_catalog
```

❶ The credentials for a user with privileges to access the given database and a JDBC URL to identify with which database you want to establish a connection

Opening and closing database connections are relatively expensive operations, so you don’t want to do that every time your application accesses data. The solution is *connection pooling*: the application establishes several connections with the database and reuses them, rather than creating new ones for each data access operation. This is a considerable performance optimization.